

Smart Metro Ticket Booking System

Requirement Analysis Phase

1. Customer Journey Map

This document walks through the full experience of a metro rider booking a ticket on the digitized ServiceNow platform, tracking every touchpoint from first hearing about the option to giving feedback after the trip — including the actions taken, how the rider feels, where the current paper-based process falls short, and what the digital solution opens up at each stage.

Figure 1.1 — Commuter journey through the digitized ticketing system, start to finish

1. Discovery

Field	Detail
Persona Involved	Passenger
User Action	Commuter finds out about digital booking through station signage, the metro app, or a ServiceNow self-service portal link shared by staff.
Emotional Response	Curious, a bit hesitant about something new
Current Pain Point (Manual Process)	No clear sign that a fast digital option even exists; long lines during busy hours.
Digital Opportunity (ServiceNow)	An in-app banner, a portal landing page, and a WhatsApp entry point make the digital route easy to spot.

2. Login / Access Portal

Field	Detail
Persona Involved	Passenger
User Action	Commuter opens the Service Portal (web/mobile) or the WhatsApp bot and logs in via mobile number/OTP, or continues as a guest.
Emotional Response	Neutral, turning positive once login is quick
Current Pain Point (Manual Process)	Manual counters need cash and a physical visit, with no saved history or reusable profile.
Digital Opportunity (ServiceNow)	OTP-based single sign-on through the Service Portal gets login done in under 30 seconds.

3. Select Station & Travel Details

Field	Detail
Persona Involved	Passenger
User Action	Commuter fills out the Service Catalog item, choosing Source, Destination, Passenger Type (General/Student/Senior Citizen), and Number of Tickets.
Emotional Response	Focused, wants things to be quick and clear
Current Pain Point (Manual Process)	Physical counters rely on verbal exchange and paper maps, so station-name mix-ups are common.
Digital Opportunity (ServiceNow)	Catalog client scripts and station reference data auto-validate routes and rule out bad combinations.

4. Fare Review

Field	Detail
Persona Involved	Passenger

Field	Detail
User Action	The system works out the fare on the spot using distance/zone logic and passenger-category discounts, then shows a clear breakup before payment.
Emotional Response	Put at ease by the transparency
Current Pain Point (Manual Process)	Fare disputes are common at counters due to unclear or manually-miscalculated pricing.
Digital Opportunity (ServiceNow)	An automated fare-logic script guarantees pricing that's consistent, auditable, and error-free.

5. Payment

Field	Detail
Persona Involved	Passenger
User Action	Commuter pays via UPI, debit/credit card, or a mobile wallet, through a Flow Designer payment sub-flow.
Emotional Response	Confident, appreciates going cashless
Current Pain Point (Manual Process)	Cash-only counters slow throughput and cause delays around making change.
Digital Opportunity (ServiceNow)	Several digital payment rails drive faster adoption and quicker transaction completion.

6. QR Ticket Generated

Field	Detail
Persona Involved	Passenger
User Action	Once payment goes through, the Flow Designer workflow immediately issues a digital ticket with a unique QR code, delivered in-app and via notification.
Emotional Response	Pleased by how quickly it's ready
Current Pain Point (Manual Process)	Paper tickets get lost, damaged, or forgotten; getting a reprint means another line.
Digital Opportunity (ServiceNow)	The QR ticket lives digitally, is always retrievable, and needs no paper at all.

7. Travel & Gate Validation

Field	Detail
Persona Involved	Passenger, Station Manager
User Action	Commuter scans the QR code at the Automated Fare Collection (AFC) gate; the Station Manager's console shows the validation status live.
Emotional Response	Trust builds from a smooth, fast entry
Current Pain Point (Manual Process)	Manual ticket checks at gates are slow and often lead to disputes over validity.
Digital Opportunity (ServiceNow)	Instant QR validation cuts gate dwell time and gives station managers real-time visibility.

8. Feedback / History

Field	Detail
Persona Involved	Passenger, Metro Operations Team
User Action	Commuter can check booking history, log a support ticket, or leave feedback; Metro Operations reviews the aggregated usage data.

Field	Detail
Emotional Response	Satisfied, feels listened to
Current Pain Point (Manual Process)	There's no structured channel today for capturing rider feedback or travel history.
Digital Opportunity (ServiceNow)	A digital trail supports data-driven calls on route optimisation and peak-time planning.

Key Takeaways

- What used to be a multi-step manual process — queue, cash, paper ticket, manual gate check — shrinks to a self-service digital flow finished in under two minutes.
- Every stage past login is now instrumented, giving Metro Operations end-to-end visibility that simply doesn't exist with paper tickets today.
- The most emotionally frustrating moments sit at the legacy touchpoints — queueing, handling cash, disputed fares — and the automated fare logic plus QR validation directly resolve all of them.
- Station Managers and IT Admins are stakeholders in this journey too — the former for live gate validation, the latter for configuring and supporting the platform.